the expression evaluates to

$$\big(\lambda_1(x)\lambda_2(y) + \lambda_1(y)\lambda_2(x)\big) \cdot 1,$$

since here $aga^{-1} \in B$ implies that $a \in B$ or a is an element of B multiplied by the transposition matrix.

If

$$g = \begin{pmatrix} x & \varepsilon y \\ y & x \end{pmatrix}, \quad x \neq y,$$

the expression on the right evaluates to 0 because matrices of this type do not have eigenvalues over $\mathbb{F}_q$ (and thus cannot be conjugated into B). From the definition, $\lambda_i(x)$ for $i = 1, 2$ is a root of unity, so

$$\begin{aligned} |G|\langle \chi_{V_{\lambda_1,\lambda_2}}, \chi_{V_{\lambda_1,\lambda_2}}\rangle &= (q+1)^2(q-1) + (q^2-1)(q-1) \\ &+ 2(q^2+q)\frac{(q-1)(q-2)}{2} + (q^2+q)\sum_{x\neq y} \lambda_1(x)\lambda_2(y)\overline{\lambda_1(y)\lambda_2(x)}. \end{aligned}$$

The last two summands come from the expansion

$$|a+b|^2 = |a|^2 + |b|^2 + a\bar{b} + \bar{a}b.$$

If

$$\lambda_1 = \lambda_2 = \mu,$$

the last term is equal to

$$(q^2+q)(q-2)(q-1),$$

and the total in this case is

$$(q+1)(q-1)[(q+1)+(q-1)+2q(q-2)] = (q+1)(q-1)2q(q-1) = 2|G|,$$

so

$$\langle \chi_{V_{\lambda_1,\lambda_2}}, \chi_{V_{\lambda_1,\lambda_2}}\rangle = 2.$$

Clearly,

$$\mathbb{C}_\mu \subseteq \operatorname{Ind}_B^G \mathbb{C}_{\mu,\mu},$$

since

$$\operatorname{Hom}_G(\mathbb{C}_\mu, \operatorname{Ind}_B^G \mathbb{C}_{\mu,\mu}) = \operatorname{Hom}_B(\mathbb{C}_\mu, \mathbb{C}_\mu) = \mathbb{C} \text{ (Theorem 5.10.1).}$$

Therefore, $\operatorname{Ind}_B^G \mathbb{C}_{\mu,\mu} = \mathbb{C}_\mu \oplus W_\mu$; W_μ is irreducible; and the character of W_μ is different for distinct values of μ, proving that W_μ are distinct.